

Overview of Linear discriminant analysis

Subject: Linear discriminant analysis

1.

Comparison to logistic regression Discriminant function analysis is very similar to logistic regression, and both can be used to answer the same research questions. Logistic regression does not have as many assumptions and restrictions as discriminant analysis. However, when discriminant analysis' assumptions are met, it is more powerful than logistic regression. Unlike logistic regression, discriminant analysis can be used with small sample sizes. It has been shown that when sample sizes are equal, and homogeneity of variance/covariance holds, discriminant analysis is more accurate. Despite all these advantages, logistic regression has none-the-less become the common choice, since the assumptions of discriminant analysis are rarely met.

2.

Discriminant Correlation Analysis (DCA) of the Haghighat article (see above) ALGLIB contains open-source LDA implementation in C# / C++ / Pascal / VBA. LDA in Python- LDA implementation in Python LDA tutorial using MS Excel Biomedical statistics. Discriminant analysis StatQuest: Linear Discriminant Analysis (LDA) clearly explained on YouTube Course notes, Discriminant function analysis by G. David Garson, NC State University Discriminant analysis tutorial in Microsoft Excel by Kardi Teknomo Course notes, Discriminant function analysis by David W. Stockburger, Missouri State University Archived 2016-03-03 at the Wayback Machine Discriminant function analysis (DA) by John Poulsen and Aaron French, San Francisco State University Archived 2011-12-15 at the Wayback Machine

3.

In the case where there are more than two classes, the analysis used in the derivation of the Fisher discriminant can be extended to find a subspace which appears to contain all of the class variability. This generalization is due to C. R. Rao. Suppose that each of C classes has a mean μ_i and the same covariance Σ . Then the scatter between class variability may be defined by the sample covariance of the class means

4.

$$\mathbf{x} \rightarrow \mathbf{T} \Sigma^{-1} (\mathbf{x} - \mu_0) = \mathbf{x} \rightarrow \mathbf{T} \Sigma^{-1} (\mathbf{x} - \mu_1) \rightarrow \Sigma^{-1/2} (\mathbf{x} - \mu_0) = \Sigma^{-1/2} (\mathbf{x} - \mu_1)$$

5.

Fisher's linear discriminant The terms Fisher's linear discriminant and LDA are often used interchangeably, although Fisher's original article actually describes a slightly different discriminant, which does not make some of the assumptions of LDA such as normally distributed classes or equal class covariances. Suppose two classes of observations have means μ_0, μ_1 and covariances Σ_0, Σ_1 . Then the linear combination of features $\mathbf{w}^T \mathbf{x}$ will have means $\mathbf{w}^T \mu_i$

and variances $w \rightarrow T \Sigma_i w \rightarrow \{\displaystyle \vec{w}\}^{\mathrm{T}} \Sigma_i \{\vec{w}\}$ for $i = 0, 1$ $\{\displaystyle i=0,1\}$. Fisher defined the separation between these two distributions to be the ratio of the variance between the classes to the variance within the classes:

6.

Marketing In marketing, discriminant analysis was once often used to determine the factors which distinguish different types of customers and/or products on the basis of surveys or other forms of collected data. Logistic regression or other methods are now more commonly used. The use of discriminant analysis in marketing can be described by the following steps:

7.

$$\Sigma_b = \frac{1}{C} \sum_{i=1}^C (\mu_i - \mu)(\mu_i - \mu)^T \quad \{\displaystyle \Sigma_b = \frac{1}{C} \sum_{i=1}^C (\mu_i - \mu)(\mu_i - \mu)^{\mathrm{T}}\}$$

8.

$$\frac{1}{2} (x \rightarrow -\mu \rightarrow 0)^T \Sigma_0 - \frac{1}{2} (x \rightarrow -\mu \rightarrow 0) + \frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} (x \rightarrow -\mu \rightarrow 1)^T \Sigma_1 - \frac{1}{2} (x \rightarrow -\mu \rightarrow 1) - \frac{1}{2} \ln |\Sigma_1| > T \quad \{\displaystyle \frac{1}{2} \{(\vec{x}) - (\vec{\mu})_0\}^{\mathrm{T}} \Sigma_0 - \frac{1}{2} \{(\vec{x}) - (\vec{\mu})_0\} + \frac{1}{2} \ln |\Sigma_0| - \frac{1}{2} \{(\vec{x}) - (\vec{\mu})_1\}^{\mathrm{T}} \Sigma_1 - \frac{1}{2} \{(\vec{x}) - (\vec{\mu})_1\} + \frac{1}{2} \ln |\Sigma_1| > T\}$$

9.

$$w \rightarrow \propto (\Sigma_0 + \Sigma_1) - 1 (\mu \rightarrow 1 - \mu \rightarrow 0) \quad \{\displaystyle \vec{w} \propto (\Sigma_0 + \Sigma_1) - (\mu_1 - \mu_0)\}$$

10.

Be sure to note that the vector $w \rightarrow \{\displaystyle \vec{w}\}$ is the normal to the discriminant hyperplane. As an example, in a two dimensional problem, the line that best divides the two groups is perpendicular to $w \rightarrow \{\displaystyle \vec{w}\}$. Generally, the data points to be discriminated are projected onto $w \rightarrow \{\displaystyle \vec{w}\}$; then the threshold that best separates the data is chosen from analysis of the one-dimensional distribution. There is no general rule for the threshold. However, if projections of points from both classes exhibit approximately the same distributions, a good choice would be the hyperplane between projections of the two means, $w \rightarrow \cdot \mu \rightarrow 0$ $\{\displaystyle \vec{w} \cdot (\vec{\mu})_0\}$ and $w \rightarrow \cdot \mu \rightarrow 1$ $\{\displaystyle \vec{w} \cdot (\vec{\mu})_1\}$. In this case the parameter c in threshold condition $w \rightarrow \cdot x \rightarrow > c$ $\{\displaystyle \vec{w} \cdot \vec{x} > c\}$ can be found explicitly:

11.

Practical use In practice, the class means and covariances are not known. They can, however, be estimated from the training set. Either the maximum likelihood estimate or the maximum a posteriori estimate may be used in place of the exact value in the above equations. Although the estimates of the covariance may be considered optimal in some sense, this does not mean that the resulting discriminant obtained by substituting these values is optimal in any sense, even if the assumption of normally distributed classes is correct. Another complication in applying LDA and Fisher's discriminant to real

data occurs when the number of measurements of each sample (i.e., the dimensionality of each data vector) exceeds the number of samples in each class. In this case, the covariance estimates do not have full rank, and so cannot be inverted. There are a number of ways to deal with this. One is to use a pseudo inverse instead of the usual matrix inverse in the above formulae. However, better numeric stability may be achieved by first projecting the problem onto the subspace spanned by Σ_b $\{\displaystyle \Sigma _b\}$. Another strategy to deal with small sample size is to use a shrinkage estimator of the covariance matrix, which can be expressed mathematically as